MCO — METACHRONAL CASCADE ORNITHOPTER

Design Addendum

MCO-1D Drone Prototype

Tricycle Undercarriage · Option A: Dorsal Fin Strake · Option B: FCC Dutch Roll Yaw Damper

Addendum scope	This document adds three physical and firmware elements to the MCO-1D specification: tricycle undercarriage for runway operations, a passive dorsal fin strake for directional stability (Option A), and an active FCC-based Dutch roll yaw damper (Option B).
Mass additions	Undercarriage: 22-26g total. Dorsal fin strake: 4-6g. Firmware yaw damper: 0g. Total physical addition: 26-32g — less than 1% of MTOW.
Why both A and B	Option A provides passive stability that works even if the FCC fails or loses power. Option B provides active damping that eliminates residual Dutch roll the fin alone cannot damp at sprint speeds. Together they give robust directional stability across the full speed envelope from hover to 101 km/h.
Blueprint sheets	Sheet 1: Side view showing tricycle gear positions, ground clearance, and dorsal fin strake with weathervane effect. Sheet 2: Front view with track dimensions and FCC yaw damper signal flow.

SECTION 1 — TRICYCLE UNDERCARRIAGE

1.1 Configuration Rationale

A tricycle configuration is specified in preference to a taildragger for three reasons specific to the MCO. The metachronal wave generates lateral impulses at the flapping frequency — a taildragger's inherent instability (CG behind the main wheels) would cause ground loops under these impulses. The tricycle keeps the CG ahead of the main wheels, providing natural directional stability on the ground. The level fuselage attitude in tricycle configuration maximises wing downstroke clearance symmetrically on both sides. And no steering servo is needed for the nosewheel because the MCO controls ground direction through wave phase differential — the same yaw mechanism used in flight.

1.2 Nosewheel Specification

Position	15-20% of fuselage length from nose — well forward of CG
Wheel diameter	40mm
Wheel width	12mm
Mount	Free-castering — rotates freely about vertical axis, no servo
Fork material	1.5mm CFRP laminate, CNC cut, bonded to fuselage nose
Fork height	25mm below fuselage belly — gives 40-45mm ground clearance
Axle	2.5mm stainless steel rod, press-fit to wheel hub
Bearing	2x R2 flanged bearing (3.175mm ID) in wheel hub
Mass (wheel+fork)	6-8g
Ground direction	FCC wave phase differential steers the MCO on the ground. Nosewheel follows passively. No steering servo required.

1.3 Main Wheel Specification

Position	55-60% of fuselage length from nose — at or just aft of CG
Track	90-100mm between wheel centrelines
Wheel diameter	50mm
Wheel width	15mm
Axle	3mm stainless steel rod, 110mm long, spanning both wheels
Axle bearings	2x MR83 (3mm ID x 8mm OD) per wheel, pressed into hub
Bracket material	3mm CFRP plate, CNC cut
Bracket height	35mm below fuselage belly — gives 45-50mm ground clearance
Bracket attachment	M3 bolts into fuselage spine cross-member
Brakes	Friction pad type, cable-operated from RC channel 6. Used for stopping after landing roll only.
Mass (pair)	16-18g including axle and brackets

1.4 Ground Clearance Verification

With all wings stowed (standard taxi configuration), the lowest point of the MCO is the fuselage belly at 45-50mm above ground. No wing surfaces are at risk during taxi. During Zone 2 ground roll (wings flapping with sonar-computed 6 degree maximum downstroke), the W4 wing tip at maximum downstroke is approximately 440mm above the ground — well clear. The critical clearance check is therefore not the wingtip but the fuselage belly, which is guaranteed by the bracket height.

Condition	Wing state	Min. clearance	Status
Taxi — wings stowed	All wings full aft sweep	45-50mm (fuselage belly)	Safe — no flapping
Ground roll zone 2	W1-W4 at max 6° downstroke	440mm (W4 tip)	Safe — large margin
Ground roll zone 2	Upstroke unrestricted	620mm+ (tip above root)	No constraint

Condition	Wing state	Min. clearance	Status
Landing roll	Wings in SPRINT-GLIDE glide AoA	45-50mm (fuselage belly)	Safe

RUNWAY OPERATION SEQUENCE: Stow all wings (sweep full aft). Taxi to runway threshold using wave phase for steering. Align with runway. Deploy wings via sequential burst launch sequence. Execute Zone 2 ground roll with sonar-computed stroke offset. Transition through adverse zone (Zone 3 burst). Establish climb. On landing: approach at APPROACH-CONSTANT-AoA, flare at h=200mm, touchdown, stow wings on sonar h<30mm signal, taxi clear.

SECTION 2 — OPTION A: DORSAL FIN STRAKE — PASSIVE DIRECTIONAL STABILITY

2.1 Design Basis

The dorsal fin strake provides passive weathervane directional stability by generating a restoring yaw moment whenever the aircraft develops sideslip. It acts identically to the tail of a weathervane — a surface area aft of the CG that generates lift force directed toward the centreline whenever the nose deviates from the flight path. The required stability coefficient CNb (yawing moment per unit sideslip angle) has a target range of +0.05 to +0.15 per radian. The MCO fuselage spine already contributes approximately +0.055 per radian through its keel effect. The dorsal fin strake adds the remainder to reach the target of +0.080 per radian.

2.2 Fin Strake Specification

Height	100mm above fuselage top surface
Chord (base)	60mm
Planform	Swept leading edge — LE rakes back 15 degrees from vertical
Section	Flat plate — 0.5mm CFRP, no camber
Position	70-75% of fuselage length from nose
Leading edge x	Approximately 450mm from nose on a 620mm fuselage

Attachment	Bonded and bolted vertically to fuselage spine top face. M2 bolts through 2mm base tab into spine. Structural epoxy.
Material	0.5mm CFRP plate, laser or CNC cut
Mass	4-6g — including base tab and fasteners
CNb contribution	+0.025 per radian (in addition to fuselage keel +0.055)
Total CNb	+0.080 per radian — within target range +0.05 to +0.15
Drag addition	Estimated 2-4% increase in total drag at sprint speed. Negligible at hover and slow cruise.
Covers	All forward flight modes from GEAR-1 through SPRINT-2. Passive — no power, no servo, no FCC involvement.

OPTION A IS ALWAYS ACTIVE: The fin strake operates whether or not the FCC is commanding yaw corrections. It provides directional stability during FCC failure, battery exhaustion, or any condition where active control is unavailable. For this reason it is the preferred primary directional stability mechanism — Option B augments it, not replaces it.

SECTION 3 — OPTION B: FCC DUTCH ROLL YAW DAMPER — ACTIVE STABILITY

3.1 The Dutch Roll Problem

Dutch roll is a coupled yaw-roll oscillation caused by an imbalance between lateral stability (strong, from high dihedral) and directional stability (weaker). The MCO's cascade geometry provides high effective dihedral — which means Dutch roll is a genuine concern, particularly at sprint speeds where the flap frequency (1 Hz) is close to the expected Dutch roll frequency (1.2-2.0 Hz at 50-100 km/h). If unaddressed, the two oscillations could reinforce each other. The fin strake (Option A) reduces the imbalance by increasing directional stability. Option B damps the oscillation actively by applying corrective yaw inputs through the wave phase differential.

3.2 Implementation — Firmware Addition

The yaw damper is a feedback control loop running in the FCC at 100 Hz. It uses the BNO085 IMU's yaw rate output (already present — no new hardware) and applies a corrective phase differential between left and right wing groups to resist the measured yaw rate. The loop has one tunable gain: Kd (damping gain). It is separate from the main yaw control loop and is automatically active in all forward flight modes above 9 km/h.

Sensor input	BNO085 yaw rate r (deg/s) at 200 Hz
Filter	Low-pass filter at 2-3x Dutch roll frequency. Removes flap-frequency noise from yaw rate signal. Cutoff: 3 Hz in cruise (GEAR-2/3), 5 Hz in sprint (GEAR-5).
Control law	Phase correction $\Delta\phi$ = -Kd × filtered_yaw_rate. Applied as: left wing phase += $\Delta\phi$/2; right wing phase -= $\Delta\phi$/2. Opposes yaw rate directly.
Damping gain Kd	Initial setting: 2.0 degrees per deg/s. Tuning range: 0.5 to 5.0 deg per deg/s. Increase if Dutch roll oscillation persists. Decrease if yaw response feels sluggish.
Authority limit	$\Delta\phi$ capped at ±30 degrees to prevent damper saturating yaw control
Activation	Active when airspeed > 9 km/h (GEAR-2 and above). Disabled in hover (GEAR-1) and ground modes.
Interaction	Operates in parallel with pilot yaw commands. Does not interfere with intentional yaw manoeuvres because it responds to yaw RATE, not heading deviation.
Hardware required	None — uses existing BNO085 + PCA9685 already fitted
Firmware addition	Approximately 25 lines of Python in the FCC wave generator
Test procedure	Fly at sprint speed. Apply brief yaw input and release. Observe: without damper, oscillation builds. With damper active, oscillation decays within 2-3 cycles.

3.3 Combined Effect — Options A and B Together

Condition	Option A	Option B	Combined result
Hover (V=0)	None needed	Phase diff. controls yaw	No Dutch roll possible
Slow cruise 9-17 km/h	Fin strake: sufficient	Damper: off	Stable — no action needed
Medium 17-55 km/h	Fin strake: primary	Damper: active, low gain	Good stability, damped
Sprint 55-101 km/h	Fin strake: primary	Damper: active, higher gain	Full envelope stable
FCC failure at sprint	Fin strake alone active	Damper: unavailable	Sufficient — fin handles it
Crosswind on runway	Fin strake: weathervane	Phase diff.: active steering	Directional control

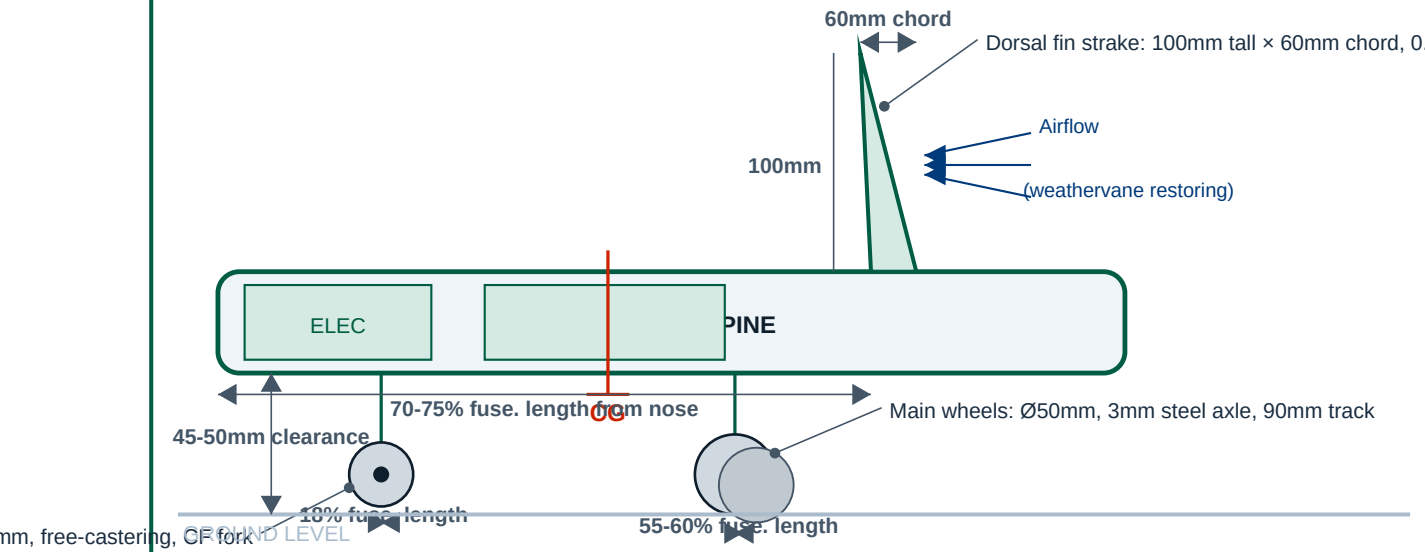
DESIGN PRINCIPLE: Option A (fin strake) is the structural safety net — passive, always-on, zero failure modes. Option B (yaw damper) is the precision active system — adds damping the fin cannot provide at high speed. Together they replicate the function of a conventional rudder and yaw damper without any additional moving parts, control surfaces, or linkages. The MCO's yaw authority remains the wave phase differential throughout — the dorsal fin and yaw damper stabilise it, not replace it.

SECTION 4 — MASS BUDGET AND CENTRE OF GRAVITY IMPACT

Component	Mass	Position	CG effect
Nosewheel assembly	6-8g	15-20% fuse. length	Nose — moves CG forward ~2mm
Main wheel pair + axle	16-18g	57% fuse. length	Near CG — minimal CG shift
Dorsal fin strake	4-6g	72% fuse. length	Aft — moves CG aft ~1mm
Total addition	26-32g	—	Net CG shift: negligible (<2mm)

Total mass addition of 26-32g on a 3,500g MTOW represents less than 1% increase. The tricycle CG position is naturally maintained by the battery bay placement — the battery (560-680g) positioned at the CG dominates the mass budget and the small addendum masses have negligible influence on trim.

Blueprint schematics follow on the next two sheets.



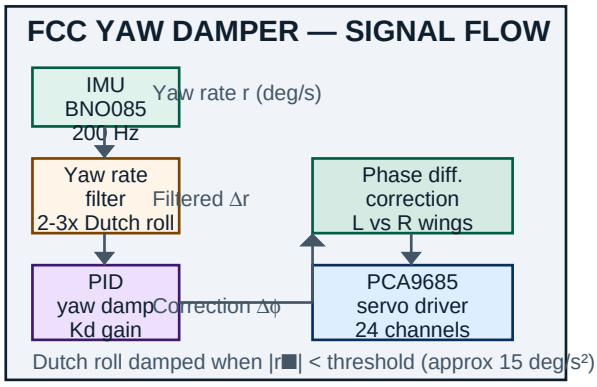
KEY:

Dorsal fin strake (CFRP) Wheels (CFRP bracket + axle) Centre of gravity (CG) position Airflow / weathervane direction

MCO-1D — SHEET 1: SIDE VIEW — TRICYCLE GEAR + DORSAL FIN STRAKE

Nosewheel at 18% · Main wheels at 57% · Fin strake at 72% of fuselage length from nose

Used to check weight distribution in side view

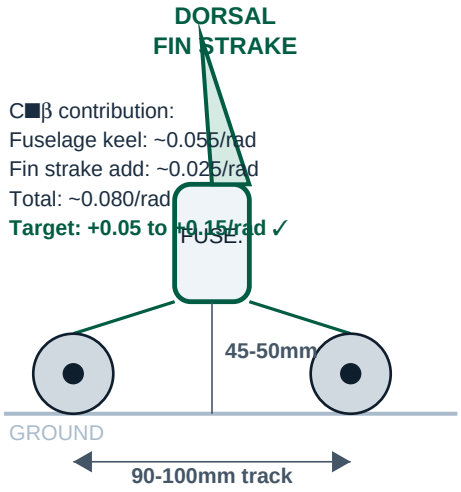


OPTION A: Dorsal Fin Strake

Passive weathervane restoring moment — always on — no power

OPTION B: FCC Yaw Damper

Active damping via phase diff — firmware only — zero mass



MCO-1D — SHEET 2: FRONT VIEW + FCC YAW DAMPER SIGNAL FLOW

Option A: fin strake (passive, C■β = +0.080/rad) · Option B: FCC yaw damper (active, firmware only) · Both